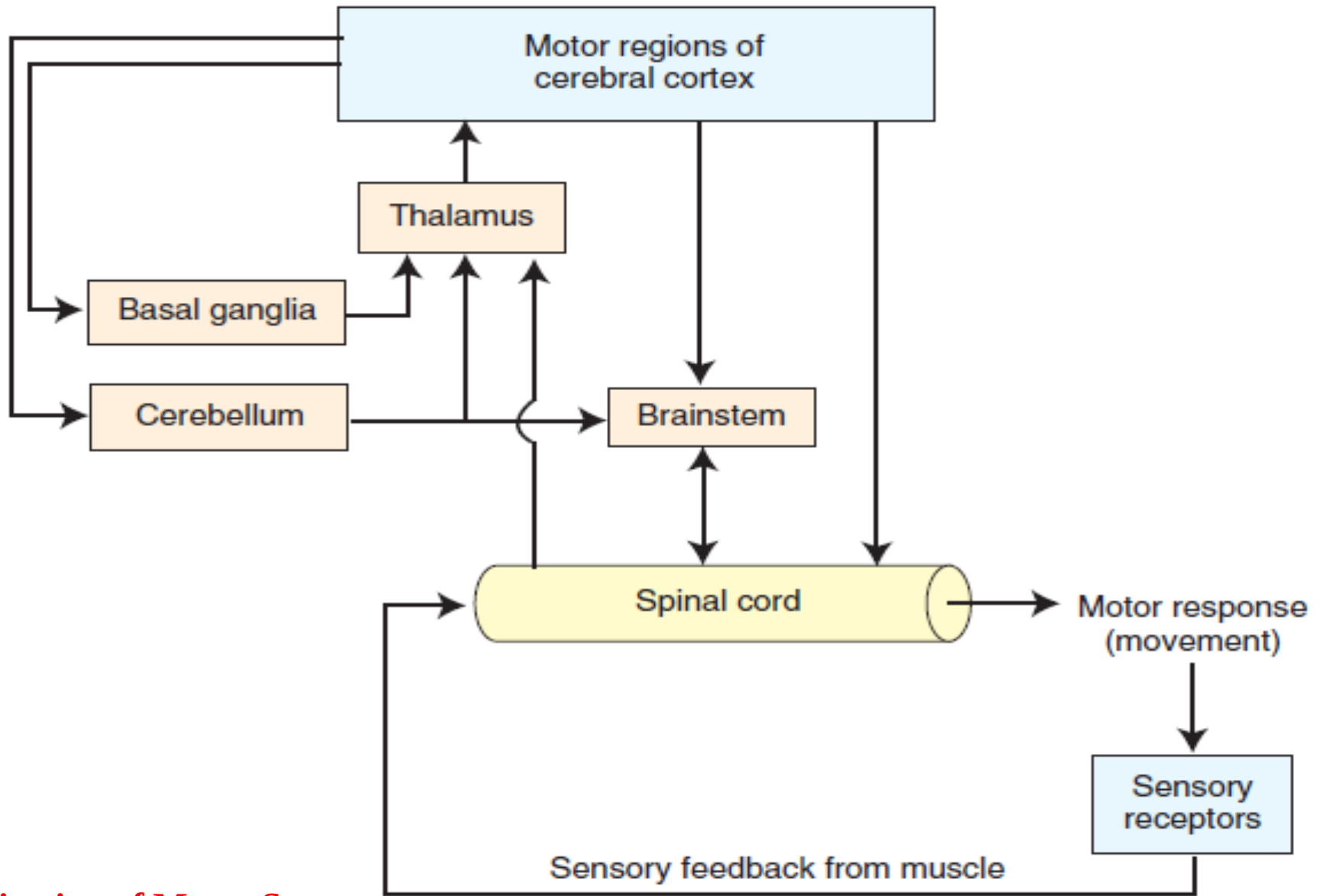
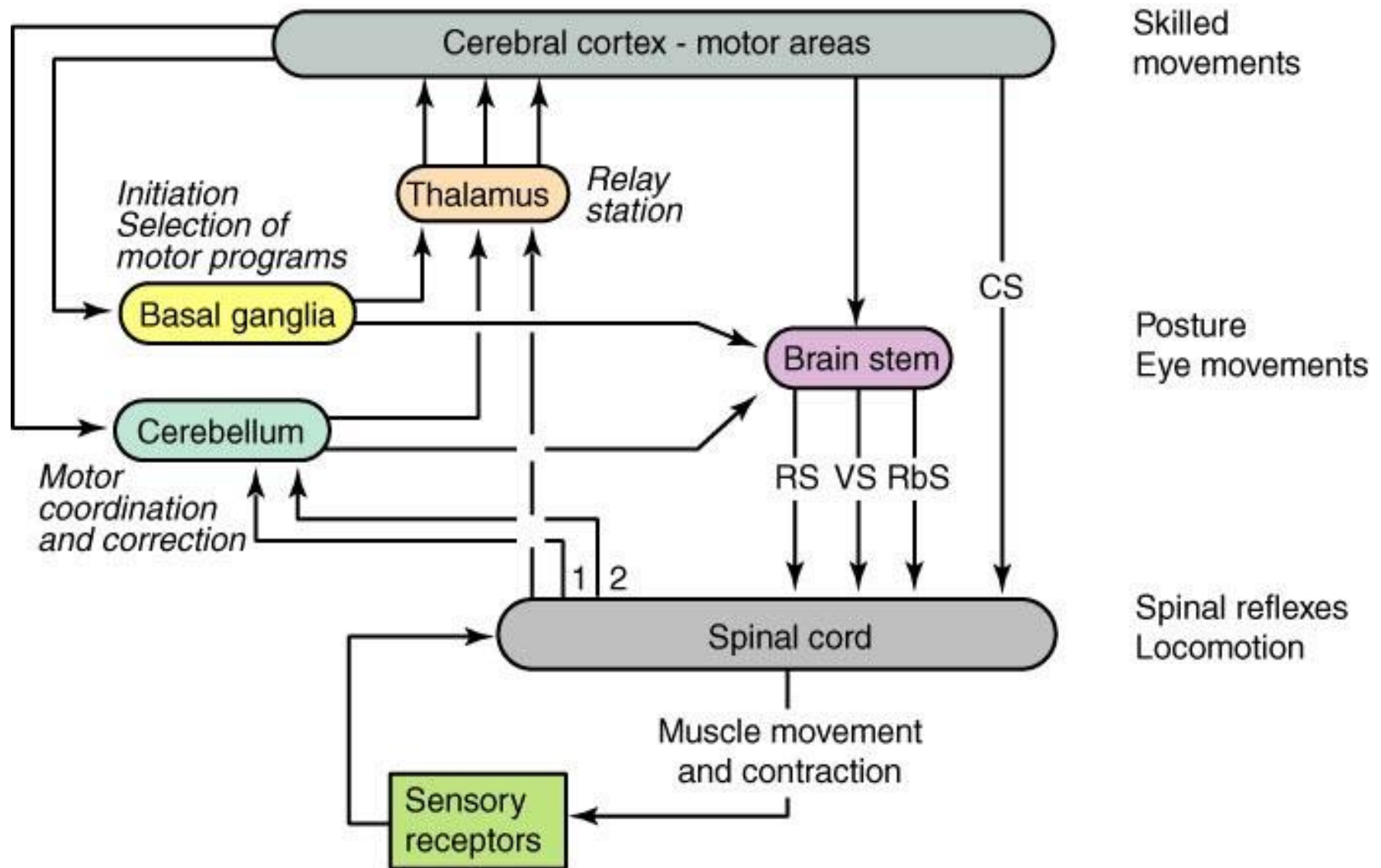

Physiology of Motor Systems

Motor Systems

- Central components:
 - Motor regions of the cerebral cortex
 - Basal ganglia + Cerebellum (+ *thalamic motor relay nuclei*)
 - Brainstem (*RF, VN, SC/IC, RN...*)
 - Spinal cord
- Peripheral components:
 - Somatic receptors (*spindles, GTOs,...*)
 - Afferent & efferent fibers
 - Effector (skeletal muscle)



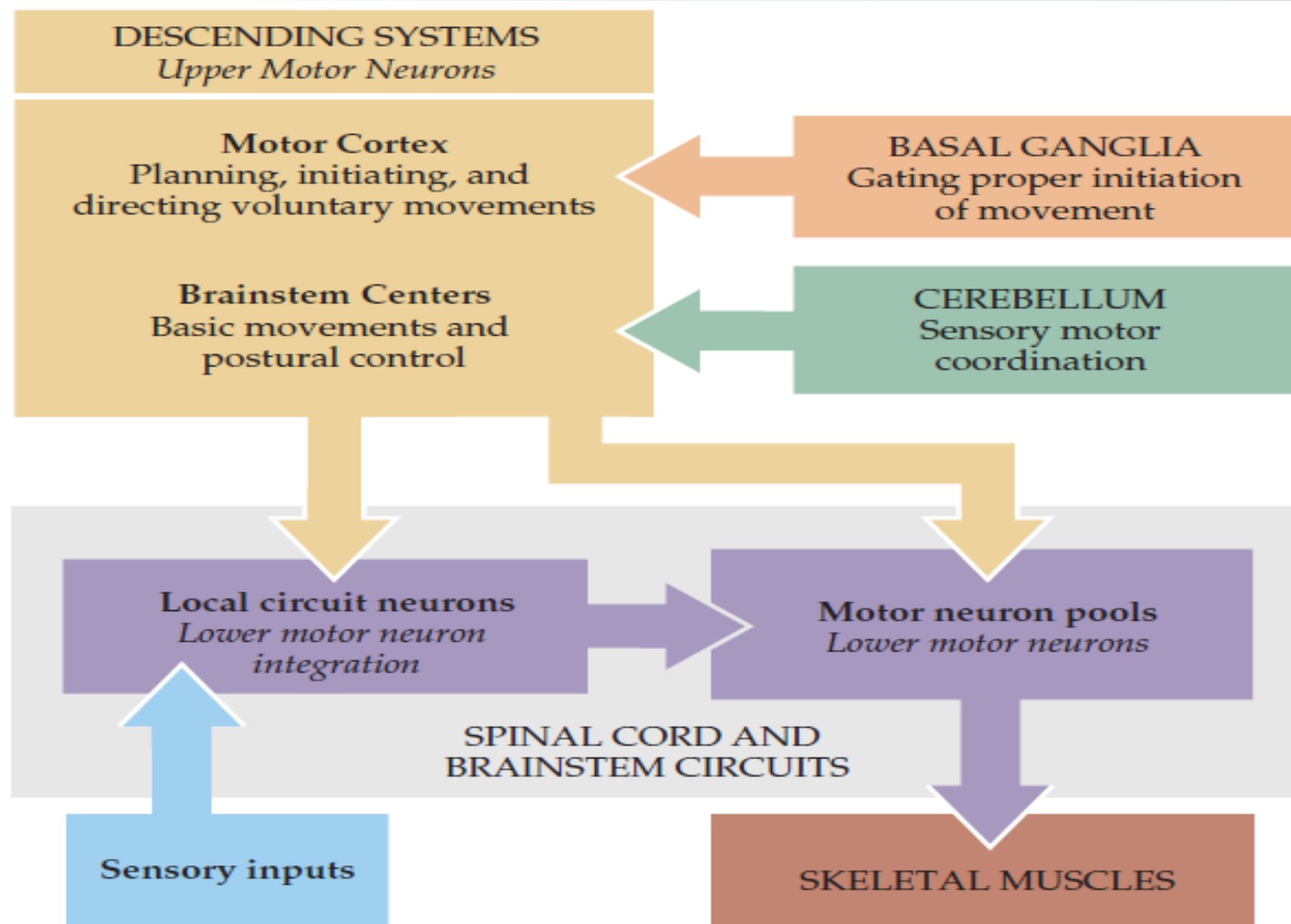
Organization of Motor Systems



Organization of Motor Systems

Neural Control of Movement

- Neural Circuits:
 - 4 interactive subsystems:
 1. Spinal cord & brainstem circuits (*LMNs + local circuit neurons*)
 2. Descending modulatory pathways (*UMNs*)
 3. Basal ganglia
 4. Cerebellum



Neural Control of Movement

1. SC & brainstem circuits:

a. LMNs:

- Their cell bodies are located in the ventral horn of the SC & in the motor nuclei of cranial nerves in the brainstem
- Innervate the skeletal muscles via the spinal peripheral nerves (SC) or via cranial nerves (brainstem)
- **NB:**
 - LMNs are the **final common pathway** for transmitting neural information from a variety of sources to the skeletal muscles

Neural Control of Movement

1. SC & brainstem circuits:

b. Local circuit neurons:

- Major source of synaptic input to LMNs
- **Inputs** → sensory + descending projections from higher centers
- Determine spatial & temporal patterns of activation of LMNs
- Mediate important sensory-motor reflexes that operate at the level of the SC & brainstem
- Provide coordination between different muscle groups that is essential for organized movement

Neural Control of Movement

2. Descending modulatory pathways:

- Comprise axons of UMNs from higher centers
- Their cell bodies are located in the cerebral cortex/brainstem
- Modulate the activity of LMNs by influencing the local circuit neurons or by directly synapsing with them
- 2 major tracts:
 - A. Pyramidal tracts
 - B. Extrapyramidal tracts

Neural Control of Movement

A. Pyramidal tracts

1. Lateral corticospinal tract
2. Anterior/ventral/medial corticospinal tract
3. Uncrossed corticospinal tracts
4. Corticonuclear/corticobulbar tract

- **Function:**

- Essential for planning, initiating and directing sequences of voluntary movements and complex spatiotemporal sequences of skilled movements (*from M-I, lateral & medial premotor cortex*)

Neural Control of Movement

B. Extrapyramidal tracts

1. Reticulospinal tract
 - Lateral/medullary (-)
 - Ventral/pontine (+)
2. Rubrospinal tract
3. Tectospinal tract
 - Lateral/SC – visuospatial reflex
 - Ventral/IC – audiospatial reflex
4. Vestibulospinal tract – M + L

Neural Control of Movement

B. Extrapyramidal tracts

- Responsible for regulating muscle tone
- Essential for control of posture and equilibrium
- Orienting the eyes, head and body with respect to vestibular, somatic, auditory, and visual sensory information
- Critical for basic navigational movements

Neural Control of Movement

3. Basal Ganglia:

- Prepare UMN circuits for the initiation of movement
- Modulate the amplitude of movement

4. Cerebellum

- Detect the difference (*motor error*) between an intended movement and the actual movement
- Thus, mediate both real-time and long-term reductions (*a form of motor learning*) in these motor errors

NB: Patients with cerebellar damage exhibit persistent **movement errors**

Neural Control of Movement

- Output pathways of the BG & cerebellum have no direct contact to either the local circuit neurons or the LMNs
- Instead, control movement by regulating the activity of UMNs

Neural Control of Movement

- **NB:**
 - **Cerebral cortex** → ultimate command (planning + modulating execution of complex voluntary motor activities)
 - **BG** → initiation of movement + modulation of motor cortex
 - **Cerebellum** → timing, speed, direction & precision of motor activity
 - **Efference copy** → plan integration + modification + coordination
 - **Brainstem** → control muscle tone, posture & equilibrium
 - **SC (LMNs)** → final execution of distinct, well-defined, voluntary activity (gait, fine movements of the hands)